Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Cross sections of solids

A survey was conducted on students' preferred modes of transportation to school. The results are summarized in the two-way frequency table below. \\ | | Bus | Bicycle | Walking | Total | |----------------|-----|---------|---------|-------| | Under 16 | 45 | 30 | 25 | 100 | | 16 and Over | 60 | 15 | 25 | 100 | | Total | 105 | 45 | 50 | 200 | Let B be the event that a randomly selected student prefers to travel by Bus. Let U be the event that a randomly selected student is Under 16 years old. \\ What is the probability that a student prefers to travel by Bus, given that they are Under 16 years old? Mathematically, this is denoted as $P(B|U)$.

2 [Foundational] • Target: Solids of revolution

A civil engineer measures the width of a river BC by setting up parallel stakes such that $DE \parallel BC$. Point D is on line AB and E is on line AC. If $AD = 45$ m, $DB = 75$ m, and $DE = 36$ m, what is the width of the river BC?

3 [Foundational] • Target: Volume of prisms and cylinders

A 5.5-foot-tall surveyor casts a 4.4-foot shadow on level ground. At the exact same moment, a nearby radio antenna tower casts a 36-foot shadow. How tall is the antenna tower?

4 [Foundational] • Target: Volume of pyramids and cones

In ΔABC, segment $DE \parallel BC$. The segment lengths are $AD = x$, $DB = x + 3$, $AE = x + 2$, and $EC = x + 7$. What is the value of x?

5 [Foundational] • Target: Volume of spheres and hemispheres

In ΔABC, ray AD bisects $\angle BAC$ with D lying on side BC. If $AB = x + 3$, $AC = 2x$, $BD = 4$, and $DC = 6$, what is the value of x?

6 [Foundational] • Target: Volume of composite solids

In an hourglass figure where $AB \parallel CD$ with intersection at E, $AE = 2x + 1$, $EC = 5x - 4$, $BE = 6$, and $ED = 9$. What is the value of x?

7 [Foundational] • Target: Cross sections of solids

In right ΔABC with altitude $CD \perp AB$, $AD = 2$ and $DB = 8$. What is the ratio of the legs $\frac{AC}{BC}$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Solids of revolution

Two similar triangles have areas of $45\ \text{cm}^2$ and $80\ \text{cm}^2$. If the perimeter of the smaller triangle is $27\ \text{cm}$, what is the perimeter of the larger triangle?

9 [Intermediate] • Target: Volume of prisms and cylinders

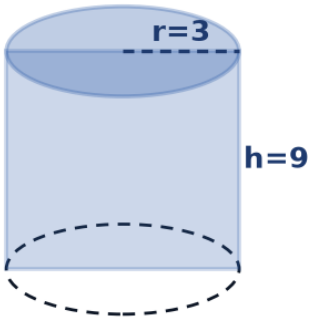
The point P has coordinates $(-3, 4)$. Point P' is the image of P after a reflection across the line $y = x$. What are the coordinates of P'?

10 [Intermediate] • Target: Volume of pyramids and cones

In right ΔABC with right angle at C and altitude $CD \perp AB$, leg $BC = 12$ and hypotenuse $AB = 16$. What is the length of segment DB?

11 [Intermediate] • Target: Volume of prisms and cylinders

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(5)$.

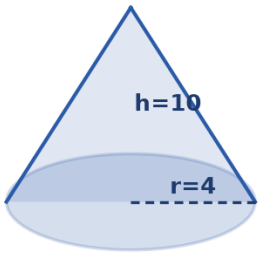


12 [Intermediate] • Target: Volume of prisms and cylinders

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

13 [Intermediate] • Target: Volume of pyramids and cones

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(2)$.



© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: Volume of pyramids and cones

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 6.01 Polygons & Quadrilaterals

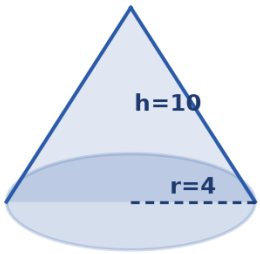
A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

16 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 7.01 Circles

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(3)$.

17 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



18 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 9.01 Transformations & Symmetry

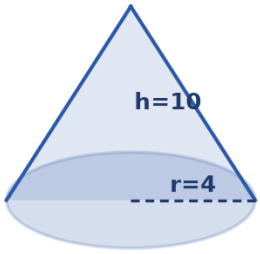
For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(4)$.

19 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 1.01 Foundations & Reasoning

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

20 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 10.01 Coordinate Geometry

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(5)$.

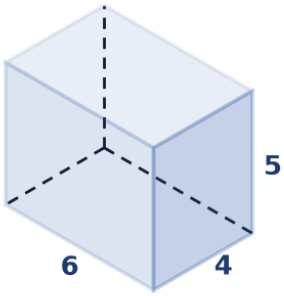


© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(2)$.



a. 12

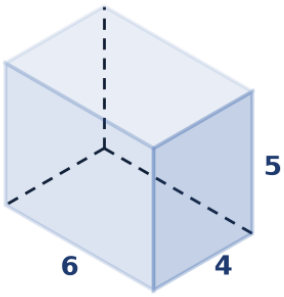
b. 15

c. 6

d. 16

22 [Foundational] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

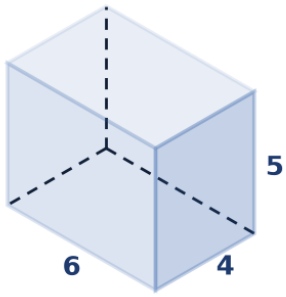
c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(5)$.



a. 21

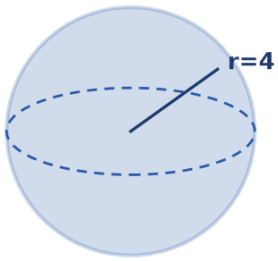
b. 24

c. 15

d. 25

24 [Foundational] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(2)$.



a. 15

b. 19

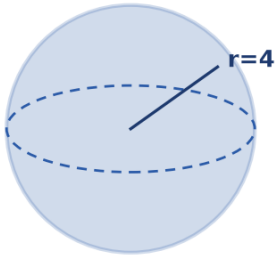
c. 8

d. 19

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

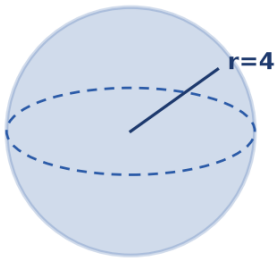
b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

26 [Foundational] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(5)$.



a. 27

b. 31

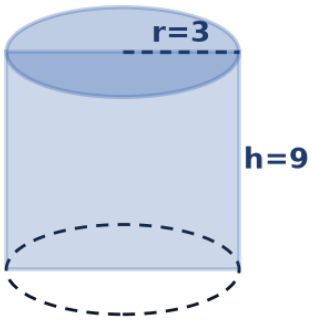
c. 20

d. 31

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Volume of prisms and cylinders

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(2)$.



a. 18

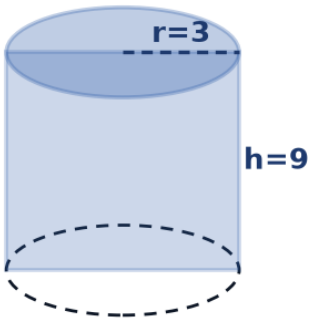
b. 23

c. 10

d. 22

28 [Foundational] • Target: Volume of prisms and cylinders

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

30 [Intermediate] • Target: Cross sections of solids

A data set recorded for cross sections of solids consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

31 [Intermediate] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(3)$.

a. 15

b. 18

c. 9

d. 19

32 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

33 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Solids of revolution

A data set recorded for solids of revolution consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

35 [Intermediate] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(3)$.

a. 19

b. 23

c. 12

d. 23

36 [Intermediate] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

37 [Intermediate] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 9.01 Transformations & Symmetry

A data set recorded for volume of prisms and cylinders consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 1.01 Foundations & Reasoning

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(3)$.

a. 23

b. 28

c. 15

d. 27

40 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 10.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

41 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

42 [Challenging] • Target: Volume of spheres and hemispheres • Cross-Lesson: 3.01 Triangles & Congruence

A data set recorded for volume of spheres and hemispheres consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

43 [Challenging] • Target: Volume of spheres and hemispheres • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

For the model $f(x) = 2x + 3$ in volume of spheres and hemispheres, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Volume of composite solids • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A data set recorded for volume of composite solids consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

45 [Challenging] • Target: Volume of composite solids • Cross-Lesson: 6.01 Polygons & Quadrilaterals

For the model $f(x) = 3x + 4$ in volume of composite solids, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

© ClipSAT — clipsat.org | All Rights Reserved